

Noah Adasi

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● EDUCATION & TRAINING

07/09/2020 - 16/08/2024 - BEREKUSO, GHANA

BSC. ELECTRICAL & ELECTRONIC ENGINEERING- ASHESI UNIVERSITY

Field(s) of study:Electrical and Electronic Engineering | **Level in EQF:** 6 | **Thesis:** Machine Learning Based Electronic Scarecrows for Protecting Agricultural Yields from Predator Damage | **Website:** <https://ashesi.edu.gh/>
Links <https://air.ashesi.edu.gh/items/57e401b2-ed70-42d6-8a82-6d6cb32ddf86> | <https://noahadasiatashesi.wixsite.com/noahadasi/course-taken>

● WORK EXPERIENCE

02/09/2024 - CURRENT - BEREKUSO, GHANA

UNIVERSITY TEACHING ASSISTANTASHESI UNIVERSITY

Faculty Intern (Teaching & Research Assistant)

- Serve as a Teaching Assistant for graduate and undergraduate courses:
 - Graduate: Mechatronics (MSc.) - Signals and Systems
 - Undergraduate: Communication Systems, Advanced Communication Systems, Digital Signal Processing, Signals and Systems, and Engineering Calculus.
- Lead study and Lab sessions, grade assignments, conduct office hours, and support Faculty with their research.

Website: <https://ashesi.edu.gh/>

01/10/2024 - CURRENT - ACCRA, GHANA

MATHEMATICS LECTUREREDUCATIONUSA – ACE CONSULT LIMITED

GRE, GMAT, and SAT Math Lecturer

- Deliver instruction in GRE, GMAT, and SAT Math to improve students' problem-solving and test performance.
- Develop tailored lesson plans and materials to cover each exam's specific math requirements.
- Monitor students' progress and adapt teaching methods to support goal achievement on standardized tests

Website: <https://educationusa.state.gov/centers/educationusa-accra-ace-consult> | **Link:** <https://drive.google.com/drive/folders/1wU3PV3m4KFSk7VTVVRv9mESOeTIT6oa?usp=sharing>

01/10/2021 - CURRENT - BEREKUSO, GHANA

AMBASSADORASHESI UNIVERSITY

Campus Ambassador

- Assist the Admissions Office with campus tours, providing prospective students & visitors with insights into Ashesi's programs and culture.
- Promote Ashesi University's mission and values through outreach and engagement activities.
- Support admissions events and presentations; collaborate with faculty and staff to offer personalized campus visits.

Link: <https://drive.google.com/file/d/1aXp2odp6pfCsy5V7yjpesJzDYltvKVgm/view?usp=sharing>

01/06/2023 - 30/08/2023 - ACCRA, GHANA

ELECTRICAL ENGINEERJOSPONG GROUPS OF COMPANIES – SEWERAGES SYSTEMS GHANA LTD

Electrical & Electronic Engineering Intern

- Installed and configured attendance tracking systems, including electronic locks for workforce monitoring.
- Performed forward and reverse motor control wiring to support operational automation.
- Assisted in troubleshooting and resolving electrical issues on-site to ensure system functionality.

Website: <https://josponggroup.com/accra-sewerage-systems-to-expand-its-operations/>

03/06/2022 - 30/08/2022 - KAMPALA, UGANDA

EDUCATIONAL RESEARCHERMUSIZI UNIVERSITY

Research Intern

- Collected and analyzed data on scholarship-providing NGOs and identified leading private and government schools in Uganda.
- Designed an Entrepreneurship & Programming curriculum for the University's Innovation Camp to enhance student skills.
- Developed student handbooks and outlined a constitution to support governance and organizational structure.

Website: <https://www.musizi.ac.ug/>

01/07/2019 - 30/09/2019 - NOBERKO- AHAFO KUKUOM, GHANA

MATH TEACHERHOPELAND INTERNATIONAL SCHOOL

Mathematics Tutor

- Taught approximately 40 JHS 3 students Core Mathematics, leading to their success in the Basic Education Certificate Examination.
- Moderated terminal examinations for 35 JHS 2 students, ensuring proper examination conduct and integrity.
- Implemented quiz programs that improved proficiency in Math and Science among students.

Website: <https://www.facebook.com/hopelandschool/>

● **PUBLICATIONS**

Design and Implementation of a Three-Phase Automatic Changeover Switch

Afriyie, A., Aho, E., Kuuchi, D., Osei, W., & Adasi, N. (n.d.). Design and Implementation of a Three-Phase Automatic Changeover Switch Using CAD Electrical Simulation.

● **CONFERENCES & SEMINARS**

01/04/2026 - 02/04/2026 Glasgow, United Kingdom

British Conference of Undergraduate Research (BCUR) 2026 Delivered oral presentations at the British Conference of Undergraduate Research (BCUR 2026) in Glasgow, UK, for the following research papers: "Signal Processing for AI Privacy in Machine Learning: Input-First Quantization-Based Training Technique" and "Machine Learning-Based Electronic Scarecrows for Protecting Agricultural Yields from Predator Damage."

Links https://drive.google.com/file/d/1-QF1VLNA42eFNUKGP3Xlyw_OepjyFsdT/view?usp=sharing |

<https://drive.google.com/file/d/12wrEWZs6mq81EmswuXRP9NgIkn1tgwFT/view?usp=sharing> |

https://drive.google.com/file/d/1WQ-indxJccykNo_bB83XlwKzH6dPunQU/view?usp=sharing

13/04/2026 - 15/04/2026 Richmond, Virginia, United States

Council on Undergraduate Research (CUR) Accepted presenter.

Selected to present research on "Assessing the Quality of Treated Drinking Water in Galamsey-Affected

Areas in Ghana” and “The Role of Climate Finance in Developing Countries: A Comprehensive Analysis of Multilateral and Blended Climate Finance for Adaptation in Africa.”

Link <https://drive.google.com/file/d/1GaAlaWacPy5gt36FtIbRtfrdGwgqHw3P/view?usp=sharing>

● **CERTIFICATIONS**

Duke University (Coursera)

Electric Industry Operations and Market (Certificates of Honor)

Link https://drive.google.com/file/d/1A33P_P2tz320mwwY_O1pLMhxEVfxYaO5/view?usp=sharing

Georgia Institute of Technology(Coursera)

Introduction to Electronics (Certificates of Honor)

Link <https://drive.google.com/file/d/1VDo2ZCoxEg7SdANI0BsGD0ai-EQXVw4y/view?usp=sharing>

21/07/2023 Global Mentorship Initiative (GMI)

Global Mentorship Initiative (GMI) Program

Selected by Ashesi University to participate in the Global Mentorship Initiative (GMI), an international professional mentorship program. Mentored by a senior industry professional from Brazil, focusing on career development, goal setting, professional planning, and leadership growth

Link <https://drive.google.com/file/d/1zVY32NMx0o6JfyUPBHCTtY11uEZf7Pol/view?usp=sharing>

● **RESEARCH PROJECTS**

01/10/2024 - 30/03/2025

AI for Health Research Fellow

African Leadership Academy South Africa

- Conducting research on the topic: “**Assessing water quality for treated drinking water in galamsey (Illegal mining) affected areas in Ghana.**”
- Collaborating with experienced healthcare research professionals for coaching and mentorship.
- Participating in structured learning modules to enhance knowledge in critical areas of medical research in Africa.

Link <https://www.instagram.com/p/DAtRYdqNsrl/>

07/01/2024 - 15/08/2024

Machine Learning Based Electronic Scarecrow for Protecting Agricultural Yields

Final Year Capstone Project

- Trained a YOLOv8 model using customized datasets to analyze real-time camera feeds for detecting/classifying bird threats.
- Designed and integrated responsive bird-scaring mechanisms.
- Created a prototype combining traditional deterrent methods with innovative technologies.

Links <https://air.ashesi.edu.gh/items/57e401b2-ed70-42d6-8a82-6d6cb32ddf86> |

<https://github.com/ADASINOAH/Machine-Learning-Based-Electronic-Scarerow-for-Protecting-Agricultural-yields-Rice-Farm->

7-Segment Display Based Digital Clock Design, Digital Systems Design

- Designed and implemented a fully functional digital clock circuit using 7-segment display logic.

Link <http://67.205.187.92/~noah.adasi/>

08/03/2024 - 29/04/2024

CNN Machine Learning Model for Option Trading

- Developed a CNN in TensorFlow to predict option trading outcomes based on financial ratios.
- Optimized model architecture through iterative adjustments to layers and nodes, enhancing predictive accuracy.
- Assessed the probability of default for new companies, facilitating data-driven decision-making.

03/11/2023 - 19/12/2023

Digital Signal Processing with Pulse Code Modulation (PCM)

- Analyzed PCM stages (sampling, quantization, encoding) to quantify effects on signal fidelity.
- Utilized MATLAB and Simulink to simulate PCM processes, validating findings and suggesting optimized bit depths

02/11/2021 - 16/12/2021

Optimization Using Gradient

- Investigated the q-gradient method, enhancing classical gradient techniques for unconstrained global optimization.
- Employed Jackson's derivative to redefine search directions and facilitate the escape from local minima.
- Conducted comparative analyses of the q-gradient method against Genetic Algorithms (GAs).

05/06/2024 - 25/07/2024

Cooperative-Spectrum-Sensing Using-Cognitive-Radio (MATLAB & Python)

- Implemented Cooperative Spectrum Sensing algorithms in MATLAB and Python to enhance channel detection accuracy.
- Conducted simulations to evaluate cognitive radio's effectiveness in dynamic spectrum management.

● **ACHIEVEMENTS AWARDS**

MasterCard Foundation Scholars Program Scholarship - Ashesi University(2020 - 2024)

Link <https://drive.google.com/file/d/1vWuWJFOt2uzWsn7pRmDUZJ6cc6AGlcM-/view?usp=sharing>

Board Member, Ashesi University's Student Council Academic Committee (2021-2023)

Link <https://drive.google.com/file/d/17g53XPy2E66ZeZptlKvf9ddL6Hy88Lt1/view?usp=sharing>

Chairperson, Electoral & Documentation committee, MasterCard Foundation-Ashesi University (2021-2022)

Ghana National Science and Math Quiz – National Level Contestant Honors, Sunyani Senior High School (2019)

Link <https://drive.google.com/file/d/1dCXapJ45FgV1spoVbplznV5CaJT5yPrh/view?usp=sharing>

● **RESEARCH INTEREST & TECHNICAL SKILLS**

Research interest & Relevant Skills

Research Interest:

Exploring applied signal processing (noise, quantization, transforms) to enhance machine learning privacy and develop efficient, privacy-preserving AI systems.

Technical Skills (Proficient):

- Programming: Python, MATLAB, C#, HTML/CSS, Arduino programming, CNC programming, PyTorch.
- AI/ML & Data Analytics: TensorFlow, Scikit-learn, YOLOv8, Data analytics using Python.
- EE & Design: Simulink, Proteus, EasyEDA, Power World, SolidWorks, Eagle, Autodesk.

● **PROFESSIONAL AFFILIATIONS**

IEEE: Student Member (No. 100767403)

Member, IEEE Young Professionals & Signal Processing Society

Founder of NPath Foundation

Focused on sustainable community development and education.

Link <https://www.linkedin.com/company/npath-foundation/posts/?feedView=images&viewAsMember=true>

- **REFERENCES**

Available upon request